

試題:

Problem 1 (15%)

Given an $n \times n$ matrix A , what is

$$\|A\|_{\infty} = \max_{\|x\|_{\infty}=1} \|Ax\|_{\infty} ?$$

That is, write $\|A\|_{\infty}$ as an expression of A 's elements.

Problem 2 (25%)

Assume A is symmetric and invertible. Formally prove that

$$\text{cond}(A) \geq 1$$

where $\text{cond}(\cdot)$ is the condition number. Assume 2-norm is used.

Hint: no need to check eigenvalues of matrices.

Problem 3 (20%)

Assume L is a lower triangular square matrix stored in the compressed row format. That is, assume arrays `val`, `col_ind`, and `row_ptr` are already available. No need to worry about declaring variables and memory allocation. Write the MATLAB code to solve

$$Lx = b$$

We further make the following assumptions and requirements:

- (1) $L_{ii} \neq 0$, for all i .
- (2) In `col_ind`, indices of each row are in ascending order.
- (3) Assume simple loops are used (It is like that you are writing Fortran/C/Java code though you use MATLAB syntax here and assume arrays start with index 1. Thus, you cannot use things such as `x(i:j)`, `find`, etc.).
- (4) The complexity must be no more than $O(\text{nnz})$.

Problem 4 (30%)

Assume a square matrix A is stored in the compressed column format and we would like to find

$$B = A^T.$$

We store B in compressed column format as well.

In the beginning, we try to find the number of non-zero elements in each column of B :

```
for i=1:n+1
    bcol_ptr(i) = 0;
nnz = acol_ptr(n+1)-1;
for i=1:nnz
    bcol_ptr(arow_ind(i)+1) = bcol_ptr(arow_ind(i)+1) +1;
bcol_ptr(1) = 1;
for i=2:n+1
    bcol_ptr(i) = bcol_ptr(i) + bcol_ptr(i-1);
```

Explain this segment of code and continue to finish the MATLAB code for obtaining B .

We further make the following assumptions and requirements:

- (1) Assume simple loops are used (It is like that you are writing Fortran/C/Java code though you use MATLAB syntax and assume arrays start with index 1. Thus, you cannot use things such as $x(i:j)$, `find`, etc.).
- (2) The complexity must be no more than $O(\text{nnz})$.
- (3) You can use only arrays `aval`, `arow_ind`, `acol_ptr`, `bval`, `bcol_ind`, `brow_ptr`. That is, you cannot allocate other arrays.

Problem 5 (10%)

Given a linear system

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix} x = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

Use $x^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ as the initial point.

- (a) Do three iterations of the Jacobi method.
- (b) Do three iterations of the Gauss-Seidel method.